

SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	Code: DSA06	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192424238 Name: Jayaprakash Nivas

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Analysis Report – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50

Expected R tasks: histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation.



Interpretation

Histogram: Shows the overall shape, center, spread, and possible skewness of the processing times.

Density plot: Gives a smoother view of the distribution and makes the concentration of values easier to see.

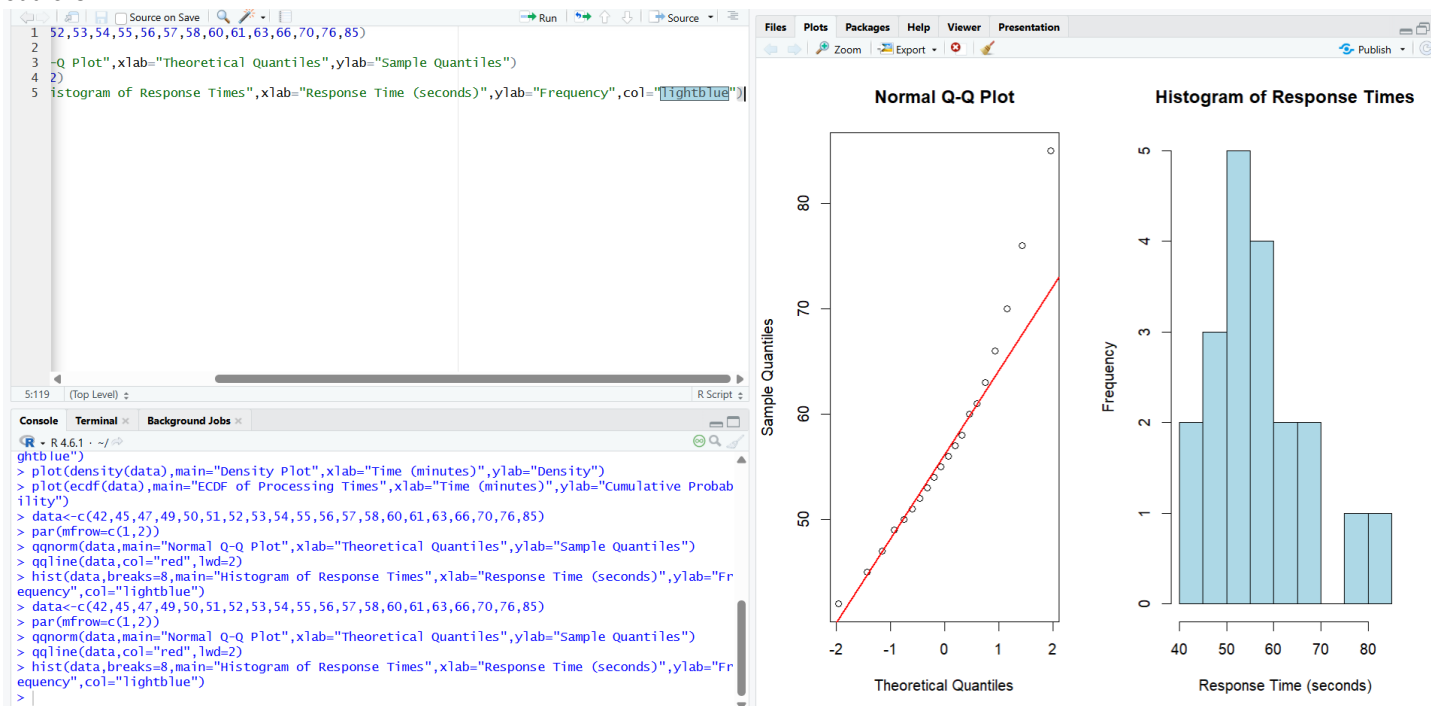
ECDF: Shows the cumulative proportion of observations at or below each processing time.

Q2. Q-Q Plot and Normality Analysis

A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.



Interpretation

Most observations lie reasonably close to the Q-Q reference line, indicating that the data are approximately normally distributed in the central region. However, the upper tail departs noticeably from the line because of the larger values, particularly 76 and 85 seconds. The histogram also shows a slight right-skewed tail. Therefore, the data are not perfectly normal and contain some possible high-end outliers.

Q3. Visualizing Many Distributions at Once

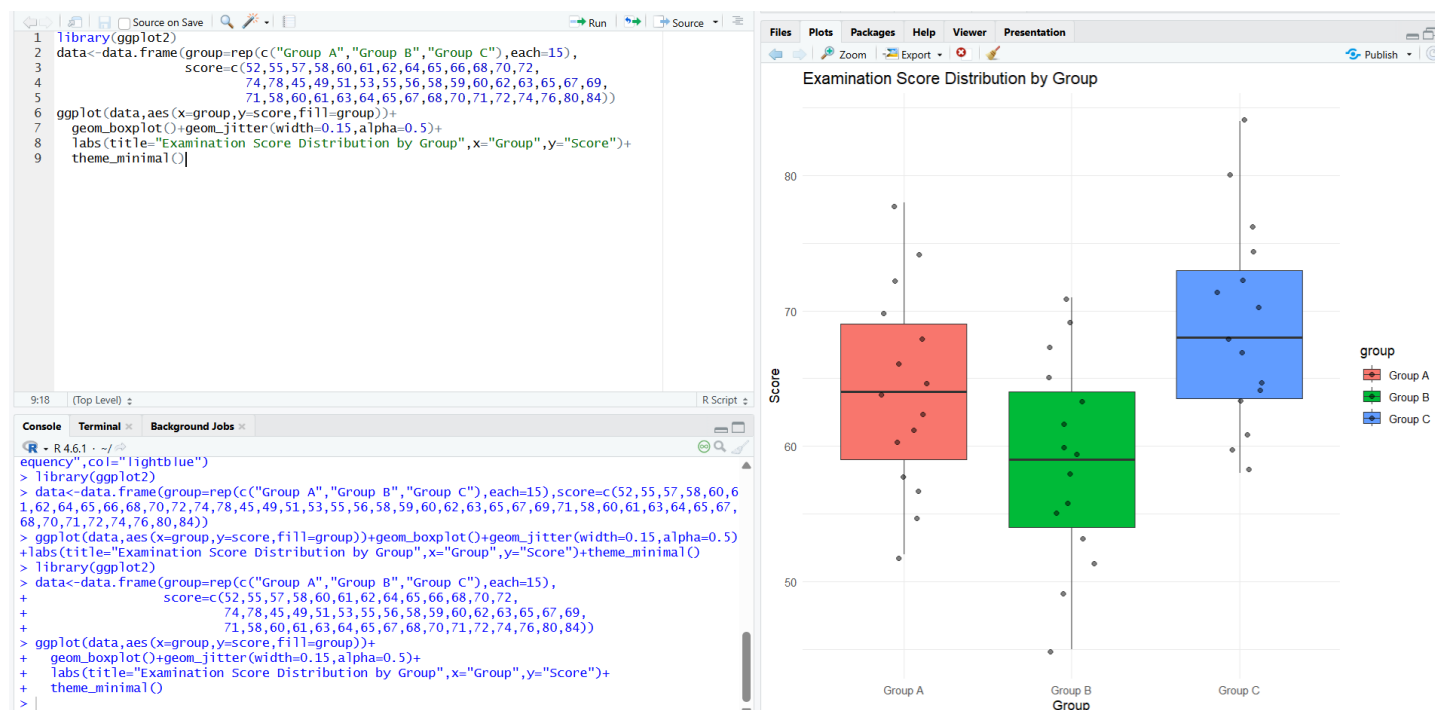
Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;

Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;

Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.



Interpretation

Group C has the highest center and generally higher scores. Group B has the lowest center, while Group A lies between Groups B and C. The boxplot allows the center and spread of all three groups to be compared easily in one figure. No major extreme outliers are apparent.

Q4. Relationships and Multivariate Visualization

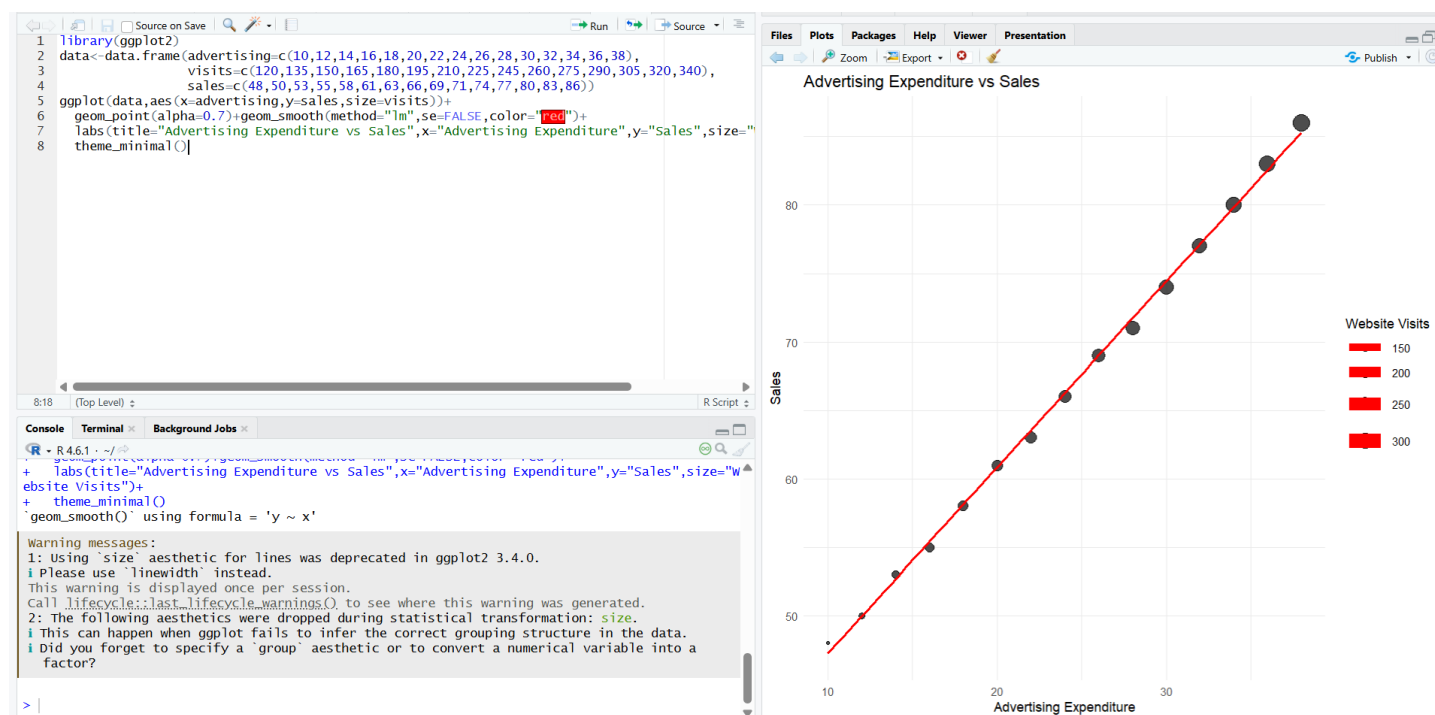
A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;

Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;

Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.



Interpretation

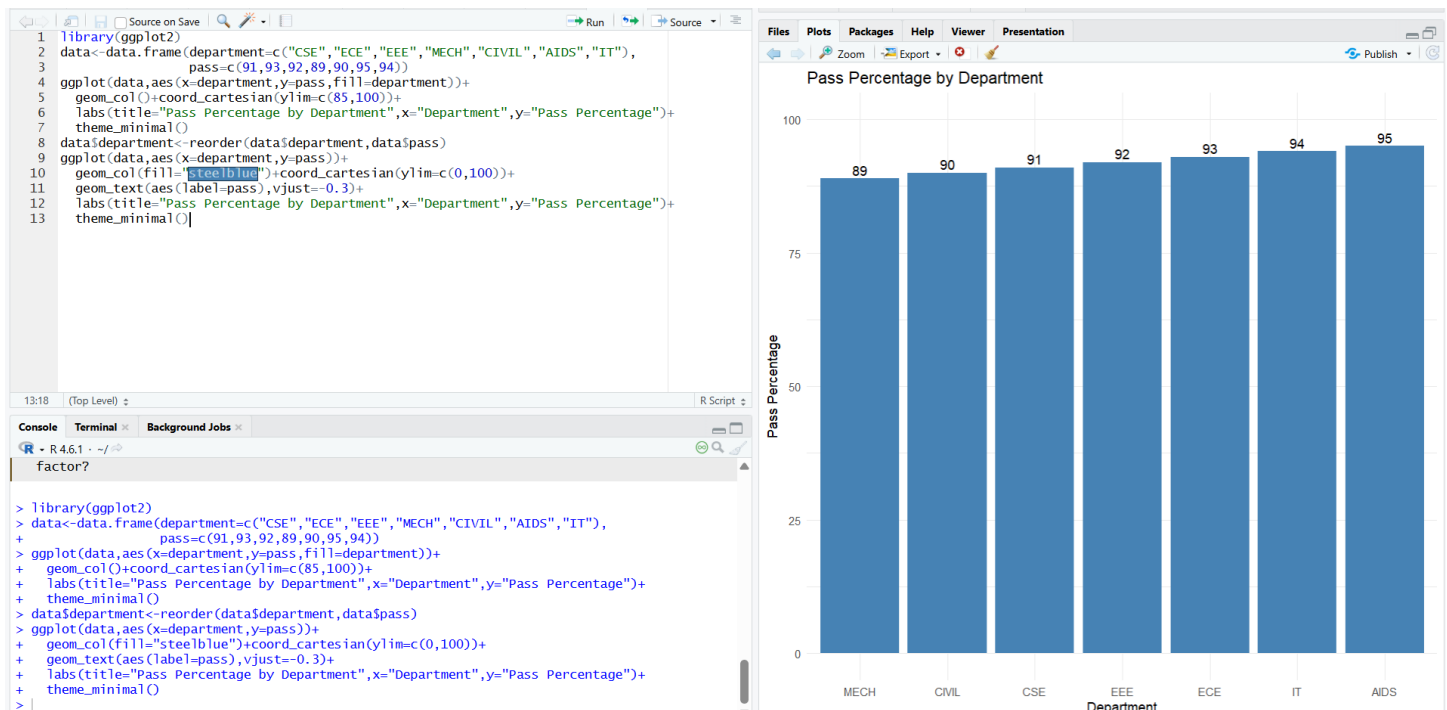
There is a strong positive relationship between advertising expenditure and sales. As advertising expenditure increases, sales also increase. Website visits add useful information because the point size represents the number of visits. However, using too many visual encodings can make the graph cluttered, so the size scale should remain clear and simple.

Q5. Misleading Visualization and Redesign

A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;
Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.



Interpretation

First plot — misleading:

The truncated y-axis starts at 85%, so the small differences between departments appear much larger than they actually are. Different colors also add unnecessary visual emphasis.

Second plot — redesigned:

The y-axis starts at zero, the departments are ordered by pass percentage, and the values are clearly labelled. This provides a more honest comparison and avoids exaggerating small differences.